P-H: a conditional recovery-work point node

15 October 2026 — proposed exact law, analytical qualification only

Decision status. P-H's causal slow-entry/recovery direction is user-approved at CP-D1. The exact reduction below is proposed at CP-D2, not yet approved or calibrated. It substitutes the compulsory point-service component of the unchanged H1/T1 predecessor. No simulator, traffic rollout, gradient computation, fitting or optimisation was performed.

1. Shared inputs and the single-component delta

Retain the predecessor's uniform footprint $[0, \ell]$, preparation length $L_p \geq \ell$, terminal length $b > 0$, $L = L_p + b$, no subsequent external arrivals, initially empty terminal, continuing symmetry $a_0 = b_0$, total mass $M = \ell(e_0 + a_0 + b_0)$, and one advice $v \in [v_-, v_+]$. Use the same positive wave speed w, jam density k_j, and triangular family

$$c(s) = \frac{wk_j s}{w+s}, \quad k_c(s) = \frac{wk_j}{w+s}, \quad r = e_0 + a_0, \quad v_+ r < R \leq c(v_-), \quad v_+ b_0 \leq R_B \leq c(v_-). \tag{1}$$

R is the fixed ordinary shared service ceiling. It is not a prescribed future receiving history. H1's paired early transfer remains

$$\dot{e} = -\frac{v}{h} \min\left\{e, \frac{k_c(v) - r + e}{\gamma}\right\}, \qquad a = r - e, \qquad b_0 \text{ constant}, \quad h > 0,\ \gamma \geq 1. \tag{2}$$

During $t_f^p = (L_p - \ell)/v < t < t_r^p = L_p/v$, terminal entrance rates are $(\dot{F}_E, \dot{F}_A, \dot{F}_B) = v(e(t), a(t), b_0)$, and zero otherwise. Define histories, extended by zero to negative times, by

$$P_j(t) = F_j(t - b/v), \quad N_j(t) = F_j(t) - Y_j(t), \quad q_j(t) = P_j(t) - Y_j(t), \qquad j = E, A, B. \tag{3}$$

N is physical stock; q is eligible unserved mass, not spatial queue length. All geometry/parameters are shared across advice values and observers.

Delta. Remove the predecessor's instantaneous extra E work factor α and the identity $Y_A = P_A$. Retain ordinary one-for-one mass admission, strict A priority and independent B service. Replace the node budget by the causal recovery history below. H1's early work factor γ, physical opportunity scale h, cutoff and control are unchanged. There is no new lateral opportunity in the terminal or spatial recovery zone.

2. An isolated restriction with a finite recovery duration

Source basis versus deduction. The point case in Leclercq et al. (2016), section 2, eq. (1), printed p4 uses bounded acceleration and backward waves to calculate cumulative passing capacity. The recovery envelope below is our isolated-case deduction, capped at the retained ceiling R. The many-entry convolution on the next page is a proposed closure, not that source's native law.

Let $m_{\mathrm{ref}} = 1$ vehicle-equivalent fix physical mass normalization. It is not a packet size or an integer admission. Define the speed at which an isolated restriction becomes neutral to R:

$$u_R = \frac{wR}{wk_j - R}, \qquad c(u_R) = R, \qquad u_R \le v_-. \tag{4}$$

Propose a representative insertion speed $0 \le u_* < u_R$ and positive recovery acceleration $a_{\mathrm{r}} > 0$. These are effective shared parameters; they are not an Eulerian exit-speed field inferred from (3). An isolated prototype at age ξ has

$$x(\xi) = u_*\xi + \tfrac{1}{2}a_{\mathrm{r}}\xi^2, \quad \theta = \xi + x(\xi)/w, \quad U(\theta) = \sqrt{(w+u_*)^2 + 2wa_{\mathrm{r}}\theta} - w. \tag{5}$$

θ is age as seen by the boundary after a backward wave, whereas ξ is prototype travel age. Equivalently, differentiating the isolated cumulative expression $wk_j(\theta - \xi(\theta))$ gives $c(U(\theta))$. Only until the restriction reaches R is that expression used.

The resulting loss profile is

$$\Theta = \frac{u_R - u_*}{a_{\mathrm{r}}} + \frac{u_R^2 - u_*^2}{2a_{\mathrm{r}}w},$$
$$D(\theta) = \begin{cases} R - c(U(\theta)), & 0 \le \theta \le \Theta, \\ 0, & \theta > \Theta. \end{cases} \tag{6}$$

D is nonnegative, decreasing, bounded by $D_0 = R - c(u_*)$, continuous at Θ and Lipschitz on $[0, \infty)$; $\|D'\|_1 = D_0$. In particular $u_* = 0$ gives $D_0 = R$, with no singularity or positive-speed epsilon. Its integrated missing work is exactly

$$W = \int_0^{\Theta} D(\theta)d\theta = \frac{k_c(u_R)(u_R - u_*)^2}{2a_{\mathrm{r}}} < \infty. \tag{7}$$

For example, use $d\theta/du = (w+u)/(a_{\mathrm{r}}w)$ and $[R - c(u)]d\theta = k_c(u_R)(u_R - u)du/a_{\mathrm{r}}$ to verify (7). Because $R, w, k_j, u_*, a_{\mathrm{r}}$ are fixed and $v \ge u_R$, this profile is fixed across advices. Advice still changes (2), arrivals and admissions; no additional direct advice-to-insertion-speed relation is asserted.

3. Proposed continuous history and conservative priority service

Constitutive assumption. Completed continuous E mass leaves mass-weighted copies of (6). Their loss demands add; the finite budget saturates at zero if those demands exhaust it:

$$p(t) = \frac{1}{m_{\text{ref}}} \int_0^t D(t-s)\, dY_E(s), \qquad \mu(t) = [R - p(t)]_+, \qquad p(0) = 0. \tag{8}$$

Only Y_E in the last Θ seconds is needed. It is predicted history, not initial hidden information, a list of individual cars, or stored longitudinal trajectories. Linear overlap of loss demands is an explicit reduction of physical interactions, not an exact wave/void calculation. Nonnegative residual service in (8) is part of this proposed service law, not an unapproved numerical clipping or recovery smoothing.

For $j = E, A$ define eligible demand almost everywhere by

$$d_j(t) = \begin{cases} c(v), & q_j(t) > 0, \\ \dot{P}_j(t), & q_j(t) = 0. \end{cases} \qquad g_A = \min\{d_A, \mu\}, \quad g_E = \min\{d_E, [\mu - g_A]_+\}, \quad \dot{Y}_j = g_j. \tag{9}$$

$Y_B = P_B$ on the retained independent B domain. At an empty queue, available arrivals restrict departure; a nonempty queue can send at $c(v)$. Since $0 \le \mu \le R \le c(v)$, a nonempty combined queue receives all available shared service. A has priority but can be delayed by (8). Exactly $g_j dt$ mass leaves, never a work-weighted amount. There is no forward escape for E or second compulsory merge of early-transferred mass.

Quantity	Units and role
$Y, F, P, N, q, m_{\text{ref}}, W$	Vehicle-equivalent mass (history, stock or integrated work).
R, c, D, p, μ, g, d	Vehicle-equivalents per second (service or demand).
u_*, u_R, v, w	Length per second.
a_{r}, Θ	Length per second squared; seconds, respectively.
p	Service-loss demand, not physical traffic stock.

For finite completed E mass M_E,

$$\int_0^\infty p(t)dt = \frac{M_E}{m_{\text{ref}}} W. \tag{10}$$

This is Tonelli's identity for the nonnegative finite-support kernel, conditional at first on the eventual admitted mass. Even before clearance, the left side is bounded above by MW/m_{ref} because $Y_E \le P_E \le M$. An E burst may have completed while p remains positive. Its influence expires by Θ after its last admission; its vehicle-time does not continue in the local objective merely because p persists.

4. Existence and uniqueness for this particular node

The queue-indicator form (9) need not be used to infer uniqueness. For a continuous candidate cumulative service $K(t) = \int_0^t \mu(s)ds$, define one-sided reflection, with histories starting at zero, by

$$\mathcal{R}(X)(t) = X(t) - \min\left\{0, \inf_{0\le s\le t} X(s)\right\}. \tag{11}$$

For fixed K, the work-conserving strict-priority solution is

$$\begin{aligned} q_A &= \mathcal{R}(P_A - K), \qquad Q = \mathcal{R}(P_A + P_E - K), \\ q_E &= Q - q_A, \qquad Y_A = P_A - q_A, \qquad Y_E = P_E + q_A - Q. \end{aligned} \tag{12}$$

Nondecreasing P_E gives $Q \ge q_A$. Checking the cases $q_A > 0$, $q_A = 0 < Q$ and $Q = 0$ recovers (9); each departure is nonnegative, nondecreasing in cumulative mass and bounded by eligible arrivals. The total rate is at most $\dot{K} \le R$, hence also respects each lane's sending ceiling. All histories are absolutely continuous.

Reflection is 2-Lipschitz in the uniform norm. Holding arrivals fixed,

$$\|\Delta Y_E\|_\infty \le 4\|\Delta K\|_\infty. \tag{13}$$

Integration by parts, using $Y_E(0) = 0$ and the absolutely continuous D on nonnegative ages, gives

$$p(t) = \frac{D_0 Y_E(t) + \int_0^t D'(t-s)Y_E(s)ds}{m_{\mathrm{ref}}}, \qquad \|\Delta p\|_\infty \le \frac{8D_0}{m_{\mathrm{ref}}}\|\Delta K\|_\infty. \tag{14}$$

Thus the causal map

$$K \;\longmapsto\; \int_0^t [R - p[Y_E[K]](s)]_+ \, ds \tag{15}$$

is a contraction on any interval of length $\eta < m_{\mathrm{ref}}/(8D_0)$ with fixed identical past histories. The closed space consists of cumulative K starting at zero with slopes in $[0, R]$. The map preserves it. Successive intervals therefore give a unique global causal solution; the same estimates and Gronwall give continuous dependence on arrivals and the kernel on finite time intervals. If $D_0 = 0$, $K = Rt$ directly. The interval length is a proof device, not a simulation time step.

No future insertion schedule or receiving trace is supplied. Continuous admissions generate (8), which determines subsequent admissions uniquely. This is a proof for (6)–(9), not an appeal to a general invariant-node theorem for an arbitrary history-dependent junction.

5. Physical storage and finite full-cohort clearance

Retain the sufficient all-action domain

$$M < H = k_j b, \qquad 0 \le \dot{F}_j \le c(v), \qquad F_j \le M. \tag{16}$$

The accepted empty-link vacancy is $k_j wt - F_j(t)$ until b/w, then $Y_j(t - b/w) + H - F_j(t)$. Before b/w, it is positive because $c(v) < k_j w$; afterwards it is at least $H - M > 0$. This proof needs only nonnegative departures, not immediate A service. H1 therefore keeps its unrestricted entrance/handoff on (16) even when either terminal queue stops temporarily. Physical stock $F_j - Y_j$ is conserved, bounded and never replaced by the service-memory variable. The domain protects H1; it does not demonstrate general spillback into the preparation window.

All traffic is eligible by $t_* = L/v$. B is then clear. Suppose combined eligible A/E mass $Q(t_*) > 0$ has not yet cleared by a time $T > t_*$. With no subsequent arrivals, Q is nonincreasing and remains positive throughout that interval. Work conservation and (8) give

$$\begin{aligned} \dot{Y}_A + \dot{Y}_E &= \mu, \qquad R \le \mu + p, \\ R(T - t_*) &\le \int_{t_*}^{T} \mu(t)dt + \int_{t_*}^{T} p(t)dt \\ &\le Q(t_*) + \frac{\overline{M}_E}{m_{\text{ref}}} W, \qquad \overline{M}_E = P_E(t_*). \end{aligned} \tag{17}$$

The last inequality uses only $Y_E \le \overline{M}_E$, not prior clearance. Letting T grow contradicts (17), proving finite clearance. In particular

$$T_{\text{clear}} \le \frac{L}{v} + \frac{Q(L/v) + (\overline{M}_E/m_{\text{ref}})W}{R}. \tag{18}$$

If $Q(t_*) = 0$, the cohort is already clear. A common, conservative horizon for all admitted advices and same-M observer comparisons is

$$\overline{T} = \frac{L}{v_-} + \frac{M + (M/m_{\text{ref}})W}{R}. \tag{19}$$

All physical and eligible queues are empty after actual clearance. Any remaining service influence is not counted as an unfinished vehicle.

Meaning. No positive service floor is inserted: $p \ge R$ can give $\mu = 0$. Finite total admitted mass creates a finite integrated burden, so temporary closure cannot persist forever. Equations (17)–(19) replace the predecessor's inapplicable positive instantaneous service bound.

6. MP6: a nonempty conditional family with A delay

This symbolic existence construction establishes possibility, not a calibrated regime, numerical scenario, or preferred advice. It can require slow recovery and a long terminal. Fix one advice $v = v_+$ satisfying (1), let $e_0, a_0 = b_0 > 0$, and choose $\gamma > 1$ so

$$\delta = k_c(v) - r > 0, \qquad (\gamma - 1)e_0 > \delta. \tag{20}$$

The receiving branch of (2) is strictly active initially. Let $L_p = \ell$, so the front enters the terminal at time zero. Measured from the first eligible arrival, at b/v, the arrival rates are

$$\lambda_A(t) = v(r - e(t)) \nearrow \lambda = vr < R, \qquad \lambda_E(t) = ve(t) > 0, \quad \lambda_A + \lambda_E = \lambda \quad (0 < t < \ell/v). \tag{21}$$

The formal extension of (2) to all positive times has $e \downarrow 0$, finite positive $\int_0^\infty e(t)dt$, and that integral scales with h. Choose h so the potential E arrivals in this extension exceed $m_{\mathrm{ref}}(R - \lambda)/D_0$.

First use $a_{\mathrm{r}} = 0$ only as an *auxiliary finite-time limit*: $D(t) = D_0 > 0$ and $\mu = [R - D_0 Y_E/m_{\mathrm{ref}}]_+$. It is not an admitted recovery law or a clearance example. While $\mu > \lambda$, all arrivals leave and Y_E accumulates the potential E arrivals. The chosen integral forces μ to reach λ at a finite time when $\lambda_E > 0$. A positive E departure rate makes $\mu < \lambda$ just afterwards. From then on μ cannot increase in this auxiliary limit. Choose a time t_1 with $\mu(t_1) < \lambda$. Since $\lambda_A \to \lambda$, eventually A demand exceeds $\mu(t_1)$ by a positive amount. If A has not already queued, its eligible stock must then start increasing. Thus $q_A(t_2) > 0$ at some finite t_2.

As $a_{\mathrm{r}} \downarrow 0$, the positive-recovery kernels (6) converge uniformly to D_0 on every fixed finite interval. The finite-interval continuous dependence from (12)–(15) preserves the strict inequality $q_A(t_2) > 0$ for sufficiently small *positive* a_{r}. Choose $\ell/v > t_2$; the finite footprint then reproduces that prefix exactly. Finally choose finite $b > M/k_j$. This changes the common eligibility delay, not the translated node argument. Initial symmetry and B's bounds can be retained, and (16)–(19) apply to the admitted positive-recovery law.

Hence the *same finite cohort* can have active early receiving limitation, positive continuing-lane waiting caused by forced-entry history, protected H1 storage and finite clearance. This establishes no empirical acceleration range, natural queue size, action split or interior optimum. The zero-recovery auxiliary limit is used only to construct a finite prefix; it is explicitly excluded from the admitted model parameter set.

7. Objective, information and sensitivity consequences

From the predecessor's uniform footprint and unchanged paired early conservation, local complete cohort time is

$$\begin{aligned} J(v) &= \int_0^{\overline{T}} [M - Y_E(t) - Y_A(t) - Y_B(t)]\, dt \\ &= \frac{M(L - \ell/2)}{v} + \int_0^{\overline{T}} (q_E(t) + q_A(t))dt. \end{aligned} \tag{22}$$

B's queue is zero on (1). This includes all pre-line waiting and the full clearance tail. The service record in (8) is not another vehicle queue. Post-line vehicle-time remains outside J even if its influence remains. The fixed bound (19) removes artificial horizon truncation; it does not hold physical arrival or queue events fixed as advice changes.

Under the retained symmetry, aggregate initial density and ending density recover $a_0 = b_0 = (\rho_0 - e_0)/2$. Shared parameters and empty initial memory then fix (2)–(3); uniqueness above fixes (8)–(9). Therefore A+E and G have exactly the same objective function for every admitted advice. At points of differentiability they also have the same derivative. This is model-relative sufficiency, not a minimum-information theorem for SUMO or a discrete-order necessity result. Unknown initial service memory would reopen G2 and is not introduced here.

The causal sensitivity path includes

$$v \;\longrightarrow\; \text{early transfer and arrival clocks} \;\longrightarrow\; Y_E \;\longrightarrow\; p \;\longrightarrow\; (Y_A, Y_E) \;\longrightarrow\; J. \tag{23}$$

The fixed prototype profile has no direct v dependence in this member. Admission times, priorities, active constraints and the recovery influence of those admissions do depend on advice. A derivative that freezes them, omits A waiting, or stops before clearance is not the complete derivative. Uniqueness and Lipschitz dependence alone do not prove differentiability at every switching event or AD/finite-difference correctness.

What is not inherited. The predecessor's positive action certificate at large α, its common-fastest parameter range, and its derivative formulas apply to its unchanged instantaneous-work member only. The old aggregate-history loading construction can still initialise the same H1 family with a newly known empty node, but a new control-information advantage requires its own objective comparison. No such advantage, positive interior optimum or useful gradient has been proved or computed here.

8. Limiting checks, physical limits and the gate

No E. If $e_0 = 0$, then $Y_E = p = 0$, $\mu = R$ and (1) admits A without waiting. The remaining complete travel time decreases with advice.

Negligible recovery burden. As $u_* \uparrow u_R$ or $a_\mathrm{r} \to \infty$, $W \to 0$. The limiting node has ordinary shared service R. It does not recover the old extra-work member with $\alpha > 1$.

Stationary connection to a simpler rival. If E departures have been constant at rate g_E over a full recovery window and the node is saturated with $p < R$, then

$$p = \frac{W}{m_\mathrm{ref}} g_E, \qquad g_A + \left(1 + \frac{W}{m_\mathrm{ref}}\right) g_E = R. \tag{24}$$

This is our stationary consequence, not the source many-event capacity formula. It explains why instantaneous work can be a serious average-flow rival. Finite-burst timing, rather than novelty of a work factor, must justify the retained history.

A delay is permitted, not inevitable. For constant A arrivals $\lambda_A < R$, empty initial memory, decreasing D and strict A priority, the barrier $p \le R - \lambda_A$ is preserved: when service reaches λ_A, E receives zero service and only recovery can change p, so it cannot rise through that barrier. A stays unqueued in this fluid case. Increasing A demand in (21) avoids that special protection. The model does not predict that every slow insertion produces a backward shockwave.

Main limits. A representative fixed insertion speed is a closure parameter, not a traffic-state-derived field. Additive loss demands and budget saturation approximate overlapping physical restrictions. No source guarantees their finite-burst fidelity. Full downstream trajectories, wave/void interactions, stochastic vehicle heterogeneity and post-line vehicle-time are not reconstructed by (8). The conservative storage domain deliberately protects H1 from terminal feedback. These limits must appear in any paper claim, parameter qualification and SUMO comparison.

Gate. Confirming CP-D2 would admit this exact law only as a conditional elementary primary model. Independent development evidence, renewed action/sensitivity analysis and the accepted SUMO S-P then S-L qualification remain. Strong aggregate-history inference, A+E, scalar search, analytical/finite-difference gradients and the phenomenological capacity-drop node are retained. No empirical law, new observation, numerical parameter set, executable evaluator or next roadmap stage is approved here.

Reference record. The accompanying source audit identifies all four papers rechecked in this pass and their access limits, including the 2011 conference version and Jin's continuous-LTM and capacity-drop interfaces. All new proofs above are for the stated proposed member. The accepted H1/T1 source/PDF are preserved unchanged.